

Earned Value Management (EVM) Professional Cheat Sheet

Comprehensive PMBOK & Industry Standard Reference for Variances, Indices, Forecasting, and Project Control Architectures

1. CORE EVM VARIABLES & DEFINITIONS

Variable	Full Name & Definition
<i>BAC</i>	Budget at Completion: Total authorized budget for the entire project scope.
<i>PV</i> (BCWS)	Planned Value: Authorized budget assigned to scheduled work to be executed up to the status date.
<i>EV</i> (BCWP)	Earned Value: Measure of work performed expressed in terms of the approved budget authorized for that work.
<i>AC</i> (ACWP)	Actual Cost: Total direct and indirect costs incurred in accomplishing the work performed (ERP ledger).

2. VARIANCE ANALYSIS (COST & SCHEDULE)

Metric & Formula	Interpretation & Decision Rule
Cost Variance (<i>CV</i>) <i>CV = EV - AC</i>	> 0: Under budget < 0: Cost overrun Quantifies exact monetary variance from actual spend.
Schedule Variance (<i>SV</i>) <i>SV = EV - PV</i>	> 0: Ahead of schedule < 0: Behind schedule Expressed in monetary value of work.

4. FORECASTING FINAL COSTS (*EAC* & *ETC*)

Forecasting Model	Formula & Application Context
EAC (Typical Variance) <i>EAC = BAC / CPI</i>	Most common model. Assumes past cost inefficiencies will continue at the same rate for remaining work.
EAC (At Budget Rate) <i>EAC = AC + (BAC - EV)</i>	Used when past variances are considered atypical and future work will be executed per original budget baseline.
EAC (Cost & Schedule) <i>EAC = AC + rac{BAC - EV}{CPI × SPI}</i>	Comprehensive model factoring both cost inefficiency and schedule delays into remaining work.
Estimate to Complete <i>ETC = EAC - AC</i>	Determines exact additional capital required from today forward to reach project completion.

5. TO-COMPLETE & COMPLETION VARIANCE

Metric & Formula	Interpretation & Governance
TCPI (vs. BAC) <i>TCPI = rac{BAC - EV}{BAC - AC}</i>	Cost performance efficiency required on <i>remaining</i> work to finish exactly on the original budget (<i>BAC</i>). If > 1.0, target is unrealistic.
TCPI (vs. EAC) <i>TCPI = rac{BAC - EV}{EAC - AC}</i>	Efficiency required on remaining work to achieve the revised forecast (<i>EAC</i>). Typically stays near current <i>CPI</i> .
Variance at Comp. <i>VAC = BAC - EAC</i>	Projected budget surplus (>0) or deficit (<0) upon final project delivery.

3. PERFORMANCE EFFICIENCY INDICES

Index & Formula	Efficiency & Threshold Rules
Cost Perf. Index (<i>CPI</i>) <i>CPI</i> = <i>EV</i> / <i>AC</i>	> 1.0: Favorable spend < 1.0: Over budget (cost inefficiency) Value earned per dollar spent.
Sched. Perf. Index (<i>SPI</i>) <i>SPI</i> = <i>EV</i> / <i>PV</i>	> 1.0: Fast progression < 1.0: Slow progression Rate of time efficiency.
Critical Ratio (<i>CR</i>) <i>CR</i> = <i>CPI</i> × <i>SPI</i>	< 0.90: Immediate PMO alarm Combined health indicator.

6. ENTERPRISE IMPLEMENTATION BEST PRACTICES

- **Cadence:** Run EVM reports monthly (or weekly for high-risk capital programs) to catch variances before compounding.
- **Integration:** Maintain strict linkage between Primavera P6 (Schedule/IMS) and ERP cost accounting actuals.
- **S-Curve Tracking:** Plot cumulative *PV*, *EV*, and *AC* to visually inspect burn-rate and acceleration patterns.
- **Avoid Early Panic:** Account for mobilization and ramp-up inefficiencies during early project phases without ignoring trends.